Reg. No									

St. Joseph's College of Arts & Science (Autonomous)

St. Joseph's College Road, Cuddalore – 607001

B.Sc DEGREE EXAMINATION - APRIL 2021

ASCS301T - STATISTICAL METHODS FOR COMPUTER APPLICATIONS - I

Time : 3 hrs **Max Marks :75**

SECTION - A (10x2 =20)

Answer ALL the questions

1. Write the scope of statistics.
2. What are the different parts of a table?
3. What is meant by Dispersion?
4. What is mean?
5. What do you understand by skewness?
6. Define the meaning of moment.
7. What is scatter diagram?
8. Write the two normal equation of Straight line.
9. Explain the term interpolation.
10. Write the uses of interpolation.

SECTION - B (5×5=25)

Answer ALL the questions

11. (a). Draw a pie diagram for the following data.

Items :	Food	Clothing	Recreation	Education	Rent	Miscellaneous
Expenditure :	87	24	11	13	25	20

(Or)

(b) Explain the different types of Classification with example.

12. (a). Calculate standard deviation for the following data.

Marks :	10	20	30	40	50	60
No. of Students :	8	12	20	10	7	3

(Or)

(b) . Calculate mean deviation from mean for the following data

X :	2	4	6	8	10
f :	1	4	6	4	1

13. (a). Calculate first moments from the following data and find out β_2 .

X :	0	1	2	3	4	5	6	7	8
f :	5	10	15	20	25	20	15	10	5

(Or)

(b) Calculate Karl pearson's co-efficient of skewness for the following data

25 15 23 40 27 25 23 25 20

14. (a). Find Karl Pearson's coefficient of Correlation from the following data

X :	12	9	8	10	11	13	7
Y :	14	8	6	9	11	12	3

(Or)

(b) From the following data calculate the co-efficient of rank correlation between X and Y

X	36	56	20	65	42	33	44	50	15	60
Y	50	35	70	25	58	75	60	45	80	38

15. (a) From the following table , estimate the value of Y when X=65 by using Newton's method of backward difference.

X	31	41	51	61	71
Y	46	66	81	93	101

(Or)

(b) From the following table interpolate the value of Y when X=1.5 by using Newton's forward method .

X	1	1.2	1.4	1.6	1.8	2.0
Y	5	6	7.5	9.5	12	16

SECTION - C (3×10=30)

Answer any THREE Questions

16. Draw less than and more than ogive and find median graphically.

Marks :	0-10	10-20	20-30	30-40	40-50	50-60
No. of students :	3	9	15	30	18	5

17. Calculate mean, median and mode for the following data.

Profit :	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50	50 – 60
No. of Shops :	12	18	27	20	17	6

18. Calculate Bowley's coefficient of skewness for the following data.

C.I :	0-10	10-20	20-30	30-40	40-50	50-60	60-70	70-80	80-90
f :	10	40	20	0	10	40	16	14	0

19. Form the two regression equations and estimate X when Y=20

X :	10	12	13	17	18
Y :	5	6	7	9	13

20. Use Lagrange's interpolation formula to estimate the weight of a 7-month –old-baby

Age(months) :	0	2	5	8	10	12
Weight(lbs) :	7.5	10.25	15	16	18	21

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